SAI SUMANTH PARUCHURI

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PROFESSIONAL SUMMARY

Software Development Engineer at Amazon specializing in production AI agents, Retrieval-Augmented Generation (RAG) and Java/AWS back ends. Built a multi-agent support bot on Amazon Bedrock that cut first-response time by 99% at 78% CSAT. 3+ years delivering Spring Boot microservices, zero-downtime cloud migrations and retrieval evaluation tooling.

SKILLS

• **AI and Agents**: Amazon Bedrock, Claude Opus 4.6, GPT-5.5, LangGraph, LangChain, AutoGen, RAG, hybrid retrieval (BM25 + dense vectors), reranking, retrieval evaluation, function calling, structured outputs, model routing, FastAPI
• **Languages**: Java, Python, JavaScript, TypeScript, SQL, C++
• **Back End**: Spring Boot, Spring Security, OAuth2, RESTful APIs, Microservices, JMS
• **Cloud and Data**: AWS (EC2, S3, Lambda, ECS, API Gateway), Docker, Kubernetes, MySQL, PostgreSQL, MongoDB, DynamoDB
• **Front End and Testing**: React, AngularJS, Node.js, Express, pytest, Jest, GitHub Actions, Git

EXPERIENCE

Amazon **Dec 2024 – Present**
Software Development Engineer *Austin, TX*

- Reduced first-response time by 99% (24+ hours to under 60 seconds) for ~90 daily support questions by architecting a multi-agent Slack bot on Amazon Bedrock with Claude Opus 4.6.
- Achieved 78% CSAT by designing a three-agent orchestration pattern: a retrieval agent for knowledge-base lookups, a critic agent enforcing multi-criteria response validation, and a self-learning agent that corrects answers daily from user feedback.
- Improved retrieval precision and reduced hallucinations by building a RAG pipeline on Open Knowledge Format (OKF) with hybrid retrieval: dense vector search, BM25 keyword matching and a reranking model.
- Reduced average data-retrieval latency by 25% across customer-facing workflows by building Java RESTful microservices with Spring Boot on AWS.
- Strengthened compliance and sustained high availability at peak traffic by integrating OAuth2 authentication with Spring Security.

Truist Bank **May 2023 – Dec 2024**
Full Stack Developer *Dallas, TX*

- Migrated 40+ REST endpoints to AWS ECS and API Gateway with zero production downtime by building cloud-native Java services.
- Cut critical user-journey load times by 5+ seconds by profiling AngularJS workflows in Chrome DevTools and adding lazy loading.
- Increased peak transaction throughput by 300+ messages per hour by building asynchronous JMS processing pipelines.

Hartex Rubber Private Limited **Sep 2020 – Aug 2021**
Software Development Intern *Hyderabad, India*

- Enabled real-time monitoring of 500+ operational entities and reduced manual data handling by building a MERN-stack application (React, Node.js, Express, MongoDB) with RESTful APIs.
- Increased code coverage and reduced post-release defects by writing unit and integration tests with Jest and Enzyme.

PROJECTS

ragscope | Open-source RAG retrieval evaluation harness (Python) github.com/saiparuchuri1118/ragscope

- Built a CI quality gate that scores RAG retrieval (hit rate, precision, recall, MRR, nDCG) against a golden dataset and fails the build on threshold or baseline regressions.
- Shipped with 184 tests at 100% line and branch coverage across Python 3.9–3.13, a CI canary that verifies the gate catches seeded bugs, and zero runtime dependencies.

EDUCATION

Southern Arkansas University **2021 – 2023**
Master's, Computer and Information Science